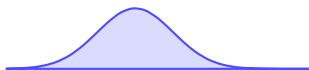
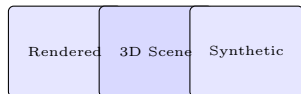


Domain Gap: Synthetic vs. Real Data Distributions

Source Domain $\mathcal{D}_S$

(Synthetic Data)



$P_S(X)$

- Perfect lighting
- Clean textures
- No sensor noise
- Idealized materials

Target Domain $\mathcal{D}_T$

(Real Data)



$P_T(X)$

- Variable lighting
- Real textures
- Sensor artifacts
- Physical materials

Domain Gap



Gap Components

- Texture differences
- Lighting variations
- Sensor noise
- Material properties
- Scene composition

$$P_S(X, Y) \neq P_T(X, Y)$$

Feature Space Representation

